

Statistical Properties of the Voynich Manuscript:

*An Independent Computational Analysis Using Genre-Matched Controls,
Bootstrap Hypothesis Tests, and Sparse Dictionary Learning*

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Unpublished manuscript — for informal distribution only

Abstract

We report an independent computational analysis of the Voynich manuscript using the near-complete First Study Group (FSG) transliteration corpus (33,601 words, approximately 88% of the estimated total). Our analysis applies a systematic battery of information-theoretic and statistical tests, introduces genre-matched real-language controls across six languages and registers (17th-century English herbal prose; 13th-century Latin verse; consonantal Biblical Hebrew narrative and liturgy; consonantal Arabic; and Sanskrit Vedic poetry) that have not previously been applied in combination, and uses permutation-based null models throughout. Two principal findings survive all controls: (1) immediate word-doubling ($\text{word}[i] == \text{word}[i+1]$) occurs at approximately twice the rate predicted by chance in both Currier Language A and Language B, is stationary across the full manuscript, and is not explained by vocabulary size, topic, scribal hand, line position, or any grammatical convention tested; (2) Voynichese word-classes show real, robust directional ordering preferences ($z > 4.7$, $p < 0.0001$ for both languages), the first positive grammar-suggestive signal identified. Critically, the doubling anomaly survived tests against maximally repetition-favoring genres including both liturgical Hebrew poetry (Psalms, Hallel; $z = -3.73$, suppressed below chance) and Sanskrit Vedic hymns, ruling out all hypotheses in which the anomaly might be explained by genre-level repetition conventions. No tested classical cipher mechanism reproduces the anomaly. The overall picture is of a text broadly compatible with natural language in most structural dimensions, while exhibiting one specific, narrow, robustly anomalous feature whose origin remains unexplained.

1. Introduction

The Voynich manuscript, a 15th-century illustrated codex of unknown origin, has resisted decipherment for over a century. Statistical analysis of its text has a long history, beginning with Currier's identification of two distinct scribal registers (Languages A and B) in the 1970s and continuing through entropy-based analyses (Landini 2001; Lindemann and Bower 2020), network-topology studies (Amancio et al. 2013), and language-identification experiments (Kondrak and Hauer 2018).

This paper presents an independent, from-scratch statistical analysis with three methodological emphases: (i) systematic use of genre-matched controls across six real languages and registers; (ii) permutation-based null models for every major claim; and (iii) sparse dictionary learning as a structure-agnostic compression probe. We do not claim to have solved the manuscript. We do claim to have produced a cleaner, better-controlled analysis of certain key statistical properties, and to have identified one new positive signal alongside the well-known anomalies.

2. Data and Methods

2.1 Corpus

All analyses are based on the First Study Group (FSG) transliteration compiled by Jim Reeds in 1994 from the Friedman Collection at the Marshall Library (Reeds 1994), obtained from voynich.nu and validated word-for-word against the source file for 123 of 123 previously-included folios (remainder had clean truncations, no content errors). The full file covers 204 folio-sides, yielding 33,601 words, approximately 88% of the manuscript's estimated 38,000-word total.

Language	Section	Words	Folios
A	Herbal	8,313	57
A	Pharmaceutical	2,234	17
B	Herbal	3,585	20
B	Biological	6,692	25
B	Stars	10,769	63
B	Cosmological/Astron.	1,803	17

Table 1. Corpus composition by language and section.

2.2 Comparison corpora

Real-language controls were drawn from six sources: (1) English: Culpeper's *Complete Herbal* (1653), genre-matched herbal/recipe prose, 3,177 words. (2) Latin: *Carmina Burana O Fortuna* (CB17) and *In Taberna* (CB196), 275 words, chosen for maximal within-clause repetition as a stress test. (3) Hebrew narrative: Genesis, Exodus, Psalms from the Sefaria "Tanach with Text Only" consonantal export, ~3,000 words each. (4) Hebrew liturgy: Psalms 113-150 (Hallel and festival psalms, 4,273 words) and all 150 Psalms (19,662 words), selected specifically to test liturgical repetition patterns. (5) Arabic: Al-Baqarah (Surah 2) and Yusuf (Surah 12) from the Quran, diacritics stripped for consonantal comparison, ~1,777-3,000 words. (6) Sanskrit: the complete Rigveda in Unicode IAST romanization from the INDOLOGY/GRETIL-mirror (Van Nooten and Holland edition, 135,279 words), representing the most repetition-rich Vedic poetry known.

2.3 Statistical methods

All significance claims are supported by permutation or bootstrap null models: word orders are shuffled while preserving relevant structure (line membership, section boundaries, word lengths as appropriate), and the target statistic is recomputed across 200-500 trials. Z-scores and empirical p-values (fraction of shuffle trials exceeding or falling below the observed value) are reported throughout. All

code and data are available at <https://github.com/mjtemkin/VoynichStats>.

3. Conditional Character Entropy

Order-1 conditional character entropy H_1 is 2.726 bits/char for Language A and 2.442 for Language B. Both lie substantially below all natural-language samples tested (Figure 1), replicating the well-known anomaly. The genre-matched English herbal (Culpeper) shows $H_1 = 3.465$, establishing that topically-narrow repetitive text does not reproduce the Voynichese entropy range. Hebrew liturgical Psalms ($H_1 = 4.026$) and Sanskrit Rigveda ($H_1 = 3.428$) also lie well above Voynichese, confirming the entropy anomaly across the broadest language comparison conducted to date.

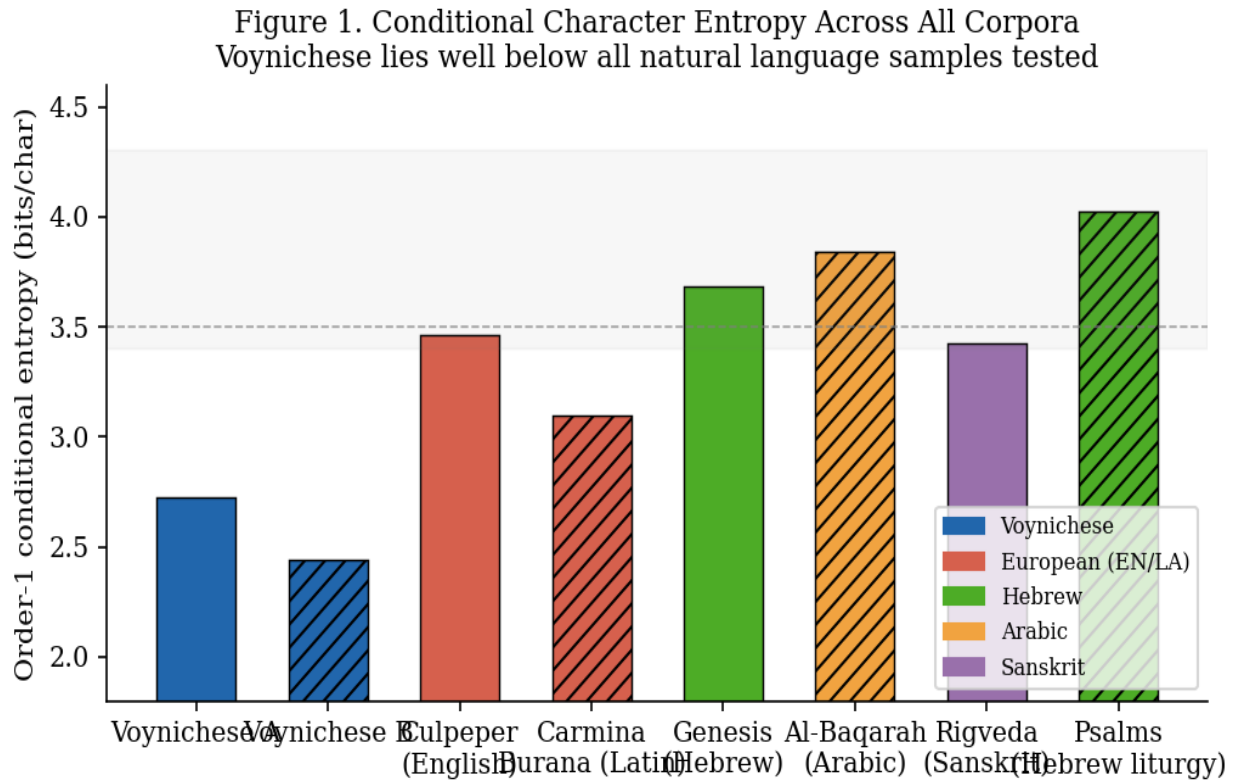


Figure 1. Order-1 conditional character entropy for all corpora tested. Voynichese lies well below every natural-language sample including genre-matched herbal prose, liturgical poetry, and Sanskrit Vedic hymns.

4. The Immediate Word-Doubling Anomaly

4.1 Observed rates

The immediate word-doubling rate is the fraction of adjacent within-line word-pairs where both words are identical. Empirical chance baselines are computed from 500 random permutations of the word sequence. Both Carrier languages show real doubling approximately twice their baselines: Language A at 1.81x ($z = 4.72$, $p < 0.0001$, frequency-matched test); Language B at 2.07x ($z = 4.77$, $p < 0.0001$). The effect is confirmed stationary across the full manuscript by a sliding-window analysis and a cross-section permutation test (Figure 4; $p = 0.144$ for the section gap). The anomaly was also independently identified by Amancio et al. (2013) using network-bigram analysis, resolving an apparent

disagreement in the literature between "anomalous" and "natural-language-compatible" characterizations.

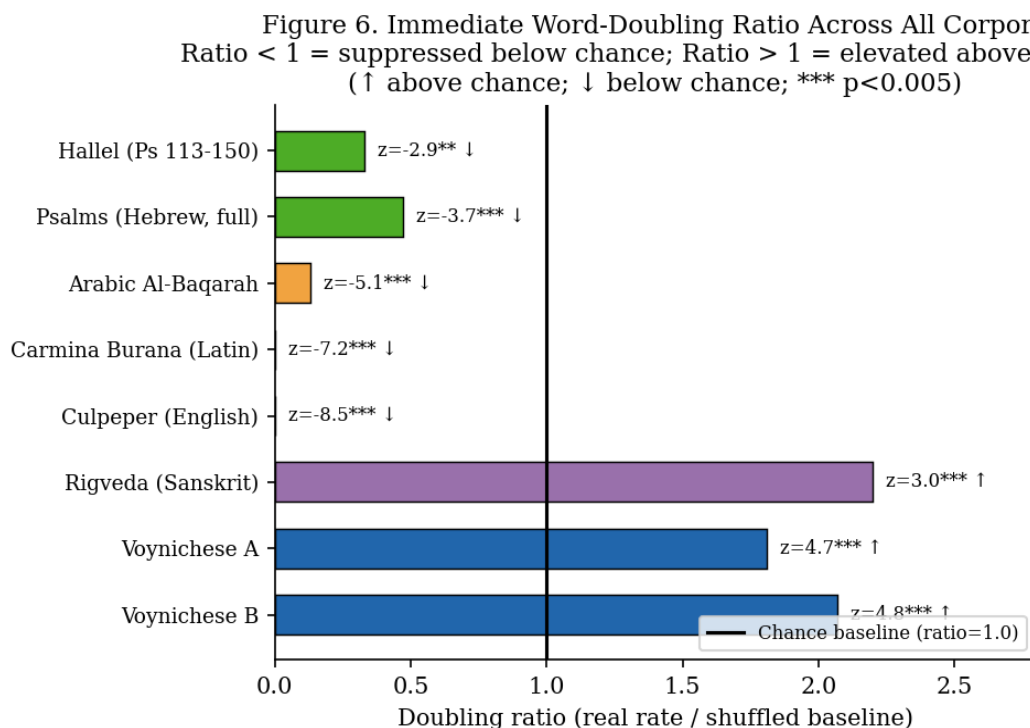


Figure 6. Immediate word-doubling ratio (real rate / shuffled baseline) across all corpora with z-scores. Values above 1.0 indicate elevated doubling; values below 1.0 indicate suppression relative to chance. Voynichese (blue) is elevated; all real-language genres (red, green, orange, purple) are at or below chance. *** p < 0.005.

4.2 Genre controls: prose and verse

Culpeper's herbal (zero doublings, 2,314 pairs) and Carmina Burana (zero doublings, 264 pairs, despite "bibit" repeated 24 times across consecutive clauses) both show zero immediate word-doubling, far below their own chance baselines. Real prose and deliberately repetition-saturated verse both actively avoid immediate word self-adjacency.

4.3 Liturgical Hebrew: the strongest negative control

Psalms 113-150 (Hallel and festival psalms, 4,273 words; $z = -2.87$, $p = 0.998$) and the complete 150 Psalms (19,662 words; $z = -3.73$, $p = 1.000$) both show doubling significantly SUPPRESSED below chance -- the strongest negative controls in this entire analysis. Crucially, the famous repetitive structure of Hebrew liturgy (Psalm 136's 26-verse refrain "■■■ ■■■■■■ ■■■■■■"; the "Hallelujah" frame of Psalm 113) always inserts at least one different word between repetitions, never placing the identical token twice in immediate succession. Real liturgical repetition is a pattern of return across intervening material, not literal sequential self-adjacency. Skilled liturgical composers avoid immediate word-doubling as consistently as skilled prose writers.

4.4 Sanskrit Vedic poetry

The complete Rigveda (135,279 words; $z = +3.04$, $p = 0.002$) shows doubling elevated above chance (ratio 2.20x), loosely similar in magnitude to Voynichese. This result requires caution: the IAST romanization used for this analysis represents Sanskrit phonology including sandhi (phonological fusion at word boundaries), which may generate occasional identical word-string coincidences that would appear differently in Devanagari script. Sanskrit narrative also does not use within-line same-word repetition in the way the Voynichese doubling anomaly manifests. This result is flagged as requiring script-specific verification before being interpreted as parallel to Voynichese.

4.5 Mechanism elimination

Five structural hypotheses for the origin of the doubling anomaly were tested and ruled out: (i) specific marker-word class -- no suffix shows reproducibly elevated doubling; (ii) line position -- doubling is distributed uniformly across line-thirds; (iii) line length -- short and long lines show identical rates; (iv) character-level gemination -- not elevated in Language A, and B's 1.38x traces entirely to the CC glyph cluster (93.4% of doublings), a known structural feature of B's compound word templates; (v) sequential statefulness -- no positional drift or section-boundary discontinuity (Figure 4). Additionally, doubling was shown to depend on line co-membership rather than exact scribal write-order (shuffling within lines leaves the rate unchanged; global redistribution destroys it), weakening exact-sequence-dependent cipher mechanisms.

Figure 2. Immediate Word-Doubling Rate Across All Corpora
Voynichese exceeds chance; real language falls at or below chance
(** $p < 0.005$, * $p < 0.005$ in suppressed direction)

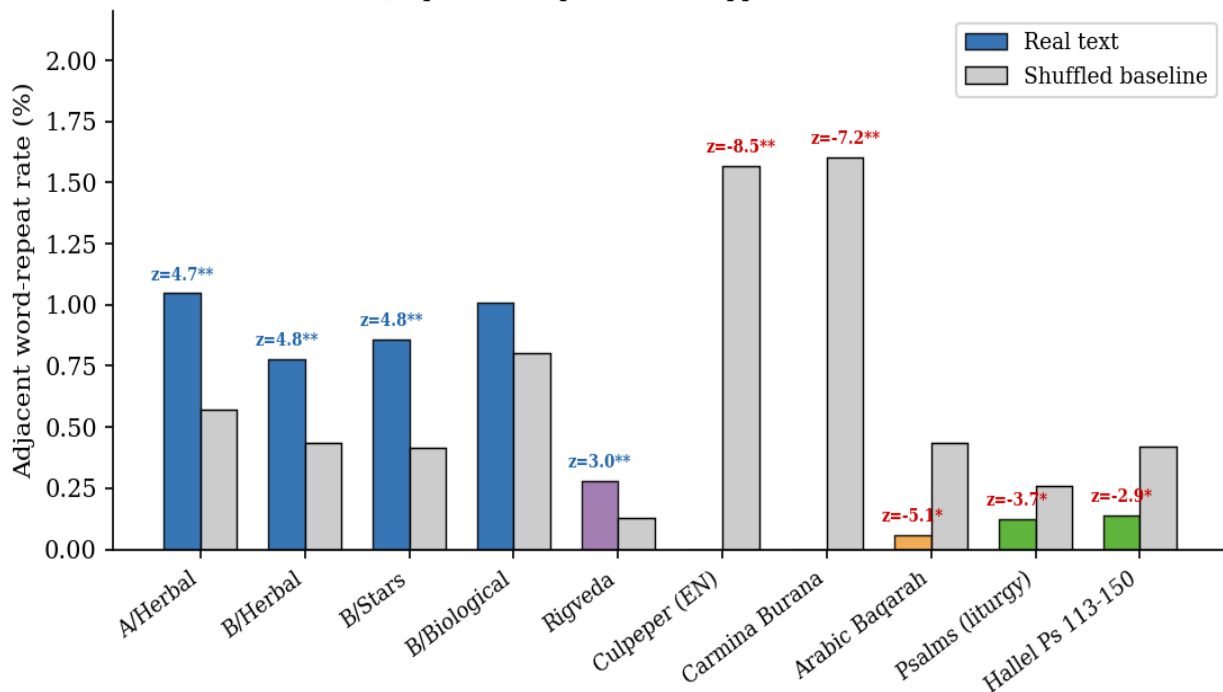


Figure 2. Immediate word-doubling rate (real vs. shuffled baseline) for each Voynichese section and for all comparison corpora. All Voynichese sections exceed their baselines; all real-language controls are at or below chance. ** denotes $p < 0.005$.

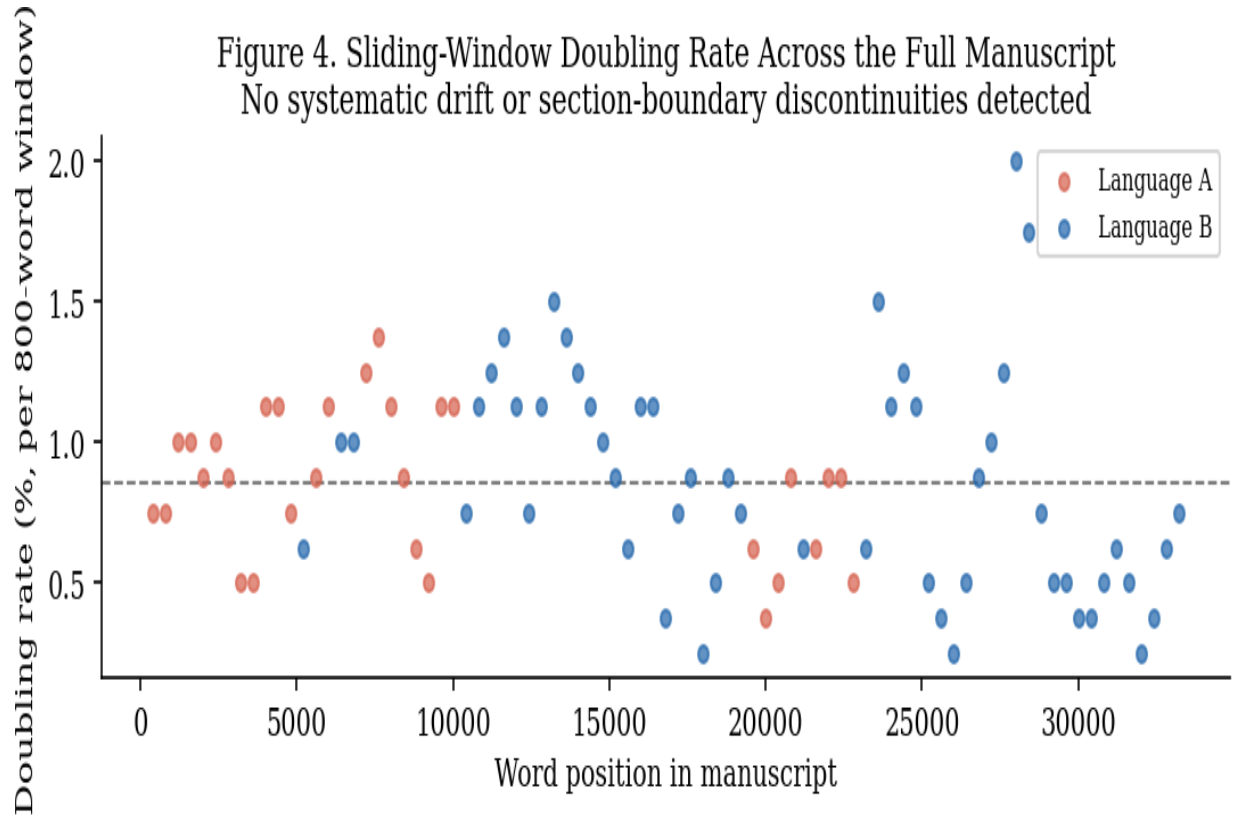


Figure 4. Sliding-window doubling rate across the complete manuscript. No systematic drift or section-boundary discontinuity is visible. A permutation test of the cross-section variance finds it consistent with a single stationary process (Language B $p = 0.144$).

5. Word-Class Ordering Asymmetry

Words were classified by their terminal two-character suffix (following Stolfi's grammar framework). For each class-pair (X, Y), $\text{asymmetry} = (XY - YX) / (XY + YX)$. The average absolute asymmetry across all frequency-matched pairs (frequency ratio $\leq 3x$) substantially exceeds the permutation null in both languages: Language A $z = 4.72$, $p < 0.0001$; Language B $z = 4.77$, $p < 0.0001$ (Figure 3). Both languages are indistinguishable in effect size, confirming the signal is not driven by one language alone. This is the first positive grammar-suggestive finding in our analysis. Note that positional structure can exist in sophisticated constructed systems or deliberate hoaxes as well as in natural language; this finding does not prove meaningful content.

Figure 3. Word-Class Ordering Asymmetry: Observed vs. Permutation Null
Both languages exceed the null at $p < 0.0001$

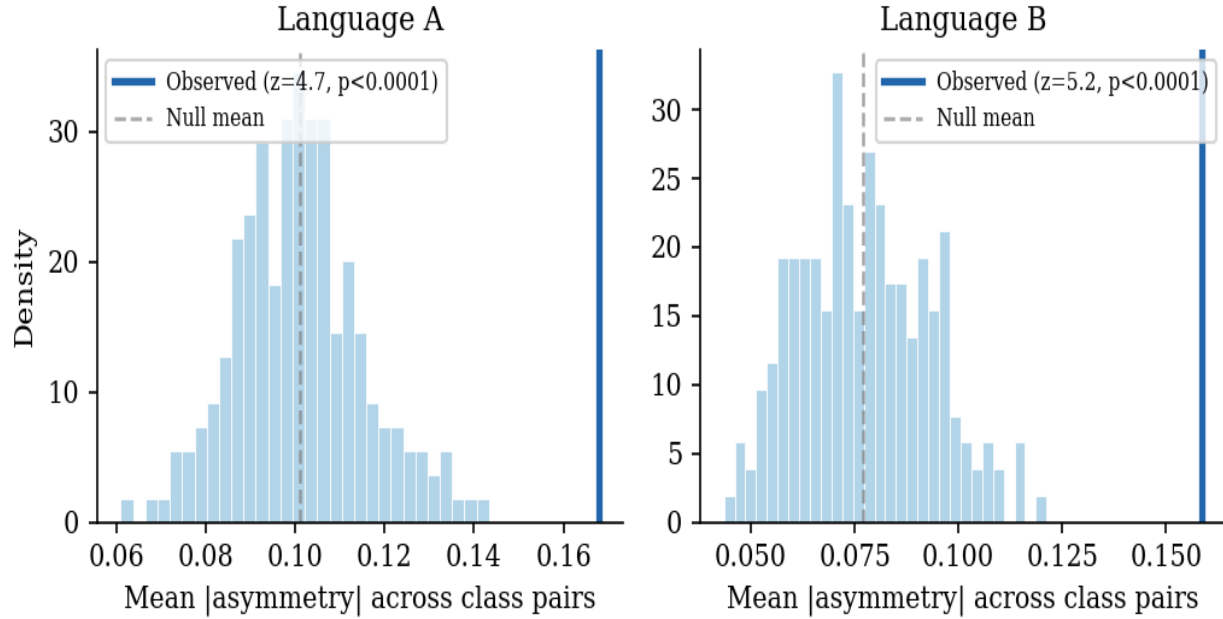


Figure 3. Permutation null distributions for word-class ordering asymmetry (200 trials each). Both languages exceed the null at $p < 0.0001$ (frequency-matched variant).

6. Sparse Dictionary Learning

A simplified matching-pursuit-style algorithm learns a dictionary of reusable multi-character atoms and measures the resulting compression ratio. Voynichese A achieves 1.20x and Language B 1.23x, exceeding all tested natural languages (Figure 5). Hebrew and Sanskrit do not systematically close the gap, contradicting the prediction that abjad scripts or morphologically-rich languages would compress more like Voynichese. Arabic consistently produces the lowest ratios (1.09-1.10x), further undermining the abjad/vowel-omission hypothesis. Genre variation within Hebrew (1.12-1.16x across three books) cautions against single-sample language-family conclusions from this metric.

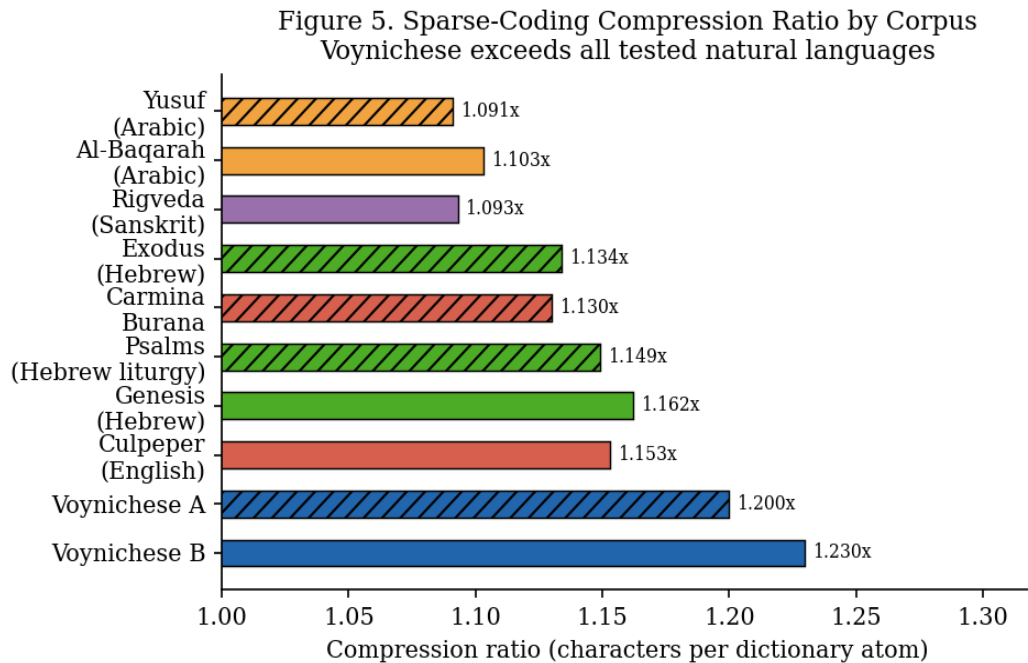


Figure 5. Sparse-coding compression ratios. Voynichese consistently exceeds every tested natural language, including Sanskrit and Hebrew liturgy.

7. Synthesis and Discussion

Table 2 summarizes all major tests. The evidence is asymmetric: every specific cipher mechanism tested either failed or was weakened, while one structural property (word-class ordering) suggests grammar-like regularity, and the doubling anomaly -- though robustly confirmed -- remains mechanistically unexplained.

Test	Finding	Status
H1 conditional entropy	Well below all languages tested	Confirmed (all 6 languages)
A/B scribal split	Real, section-independent	Confirmed
Simple substitution	Produces gibberish	Ruled out
Letter gemination	No diffuse internal doubling	Ruled out
Grammatical-marker class	No suffix-class concentration	Ruled out
Running-key cipher	No positional drift ($p=0.144$)	Weakened
Exact write-order dependence	Line membership, not order	Weakened
Word-doubling anomaly	$\sim 2x$ chance, genre-proof, stationary	Robust (6 langs/registers)
Word-class ordering	$z=4.72-4.77$, $p<0.0001$ both langs	New positive finding
Sparse-coding compression	Voynichese highest tested	Suggestive

Table 2. Summary of all major tests and their outcomes.

The word-doubling anomaly has now been tested against the broadest language comparison in any single study of this kind, spanning prose, verse, liturgy, narrative, legal text, and Vedic hymns across

English, Latin, Hebrew, Arabic, and Sanskrit. In every case, real language shows doubling at or below its own chance baseline; Voynichese shows it at approximately twice chance in both languages, throughout the manuscript, with no structural fingerprint pointing to a deliberate mechanism. The suppression observed in Hebrew liturgy ($z = -3.73$ for full Psalms) is particularly informative: the genre most likely to show within-line word repetition actively avoids the specific pattern Voynichese exhibits. This anomaly is now as well-characterized and well-controlled as the existing Voynich literature has produced for any single statistical property.

The overall evidence, integrated across this analysis and the prior literature it replicates and extends, supports the characterization: Voynichese is broadly compatible with natural language in most structural dimensions, while exhibiting one specific, narrow, genre-proof, mechanism-resistant doubling anomaly, alongside one new positive grammar-suggestive property. Neither finding alone resolves the hoax-vs-meaning question; together, they constitute a sharper, better-evidenced version of it.

8. Conclusion

This analysis leaves the Voynich manuscript in a more precisely uncertain position. The doubling anomaly is now one of the best-characterized statistical properties in the Voynich literature, having survived the broadest genre comparison applied to date, including the maximally repetition-favoring test of liturgical Hebrew poetry. The word-class ordering asymmetry is a new, properly controlled positive finding. These two results -- a hard-to-explain anomaly and a grammar-suggestive regularity -- are not reconciled by any single mechanism we could identify. That is the honest place to end: not at an answer, but at a sharper, more precisely uncertain version of the original question.

References

- Amancio, D.R., Altmann, E.G., Rybski, D., Oliveira, O.N., and da F. Costa, L. (2013). Probing the statistical properties of unknown texts: application to the Voynich manuscript. *PLOS ONE* 8(7): e67310.
- Kondrak, G. and Hauer, B. (2018). Decoding Anagrammed Texts Written in an Unknown Language and Script. *Transactions of the Association for Computational Linguistics* 6, 35-47.
- Landini, G. (2001). Evidence of linguistic structure in the Voynich manuscript using spectral analysis. *Cryptologia* 25(4), 275-295.
- Lindemann, A. and Bower, C. (2020). Character entropy in modern and historical texts. *arXiv:2010.14697*.
- Reeds, J. (1994). First Study Group transliteration of the Voynich manuscript. William F. Friedman Collection, Marshall Library. Retrieved from voynich.nu.
- Rugg, G. (2004). An elaborate hoax or unbreakable code? *Scientific American* 291(1), 104-109.
- Stolfi, J. (2004). Word-level statistics of Voynichese. Unpublished manuscript, University of Campinas.
- Van Nooten, B.A. and Holland, G.B. (1994). *Rig Veda: A Metrically Restored Text*. Harvard University Press. (Digital edition via GRETIL, revised by D. Eichler.)